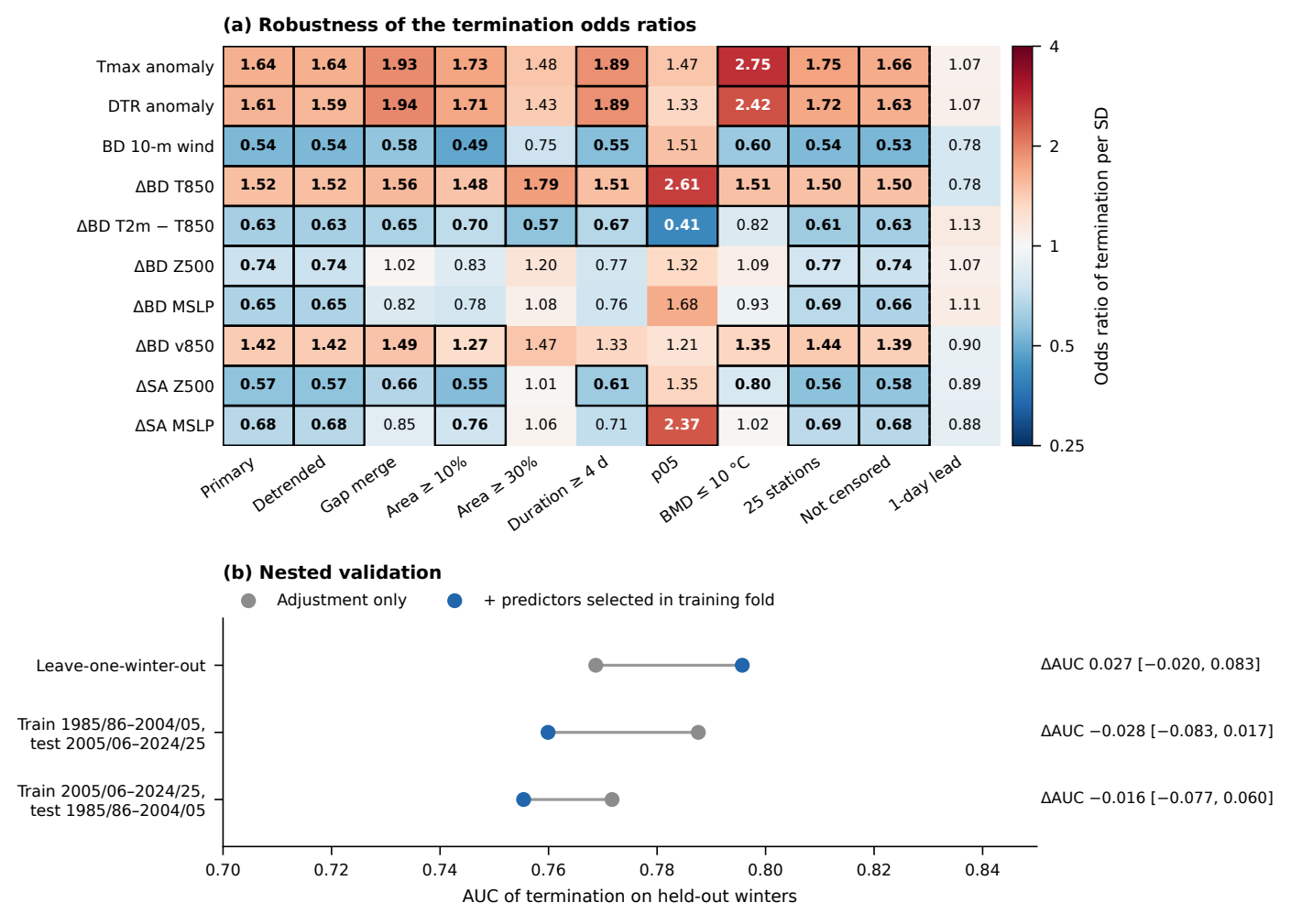

(a) Robustness of the termination odds ratios
Tmax anomaly 1.64 1.64 1.93 1.73 1.48 1.89 1.47 2.75 1.75 1.66 1.07
DTR anomaly 1.61 1.59 1.94 1.71 1.43 1.89 1.33 2.42 1.72 1.63 1.07
BD 10-m wind 0.54 0.54 0.58 0.49 0.75 0.55 1.51 0.60 0.54 0.53 0.78
ΔBD T850 1.52 1.52 1.56 1.48 1.79 1.51 2.61 1.51 1.50 1.50 0.78
ΔBD T2m − T850 0.63 0.63 0.65 0.70 0.57 0.67 0.41 0.82 0.61 0.63 1.13
ΔBD Z500 0.74 0.74 1.02 0.83 1.20 0.77 1.32 1.09 0.77 0.74 1.07
ΔBD MSLP 0.65 0.65 0.82 0.78 1.08 0.76 1.68 0.93 0.69 0.66 1.11
ΔBD v850 1.42 1.42 1.49 1.27 1.47 1.33 1.21 1.35 1.44 1.39 0.90
ΔSA Z500 0.57 0.57 0.66 0.55 1.01 0.61 1.35 0.80 0.56 0.58 0.89
ΔSA MSLP 0.68 0.68 0.85 0.76 1.06 0.71 2.37 1.02 0.69 0.68 0.88
Primary
Detrended
Gap merge
Area ≥ 10%
Area ≥ 30%
Duration ≥ 4 d
p05
BMD ≤ 10 °C
25 stations
Not censored
1-day lead
Odds ratio of termination per SD
4
2
1
0.5
0.25
(b) Nested validation
Adjustment only
+ predictors selected in training fold
Leave-one-winter-out
Train 1985/86–2004/05, test 2005/06–2024/25
Train 2005/06–2024/25, test 1985/86–2004/05
ΔAUC 0.027 [−0.020, 0.083]
ΔAUC −0.028 [−0.083, 0.017]
ΔAUC −0.016 [−0.077, 0.060]
0.70
0.72
0.74
0.76
0.78
0.80
0.82
0.84
AUC of termination on held-out winters